RAKESH CHOWDARY POLINA

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PROFILE

Experienced Software Engineer with a proven track record in developing scalable microservices, implementing robust facial recognition pipelines, and delivering cloud-native solutions. Skilled in Python, Java, Spring Boot, Flask, React.js, and AWS, with expertise in containerization, CI/CD, and Agile methodologies. Adept at optimizing performance, enhancing user engagement, and driving operational efficiency in production environments.

EDUCATION

Master of Science in Computer Science **May 2024**
Oklahoma State University, Stillwater, OK **CGPA: 3.909/4**

SKILLS

Languages:	Java, JavaScript, Python, SQL, C, C++, HTML, CSS, Swift
Frameworks:	React JS, Angular JS, Spring boot, Node JS, Hibernate, Spring MVC, React Native
Databases:	MySQL, MongoDB, NoSQL, PostgreSQL, Cassandra
Web Technologies:	RESTful API, Spring JDBC, JPA, OAuth 2, jQuery, JSP, JMS, RabbitMQ
Web Services:	REST, SOAP
Tools/Platforms:	Docker, Git, JIRA, Jenkins, Postman, Ansible, Kubernetes, Apache Kafka, Terraform, Swagger
AWS Cloud Services:	EC2, S3, EKS, RDS, ELB, Cloud Watch, CloudFront
Build Tools:	Maven, Gradle

WORK EXPERIENCE

Research Data Analyst, Oklahoma State University **May 2023 – May 2024**

- Engineered and implemented a robust facial recognition pipeline, performing both quantitative and qualitative analysis on image data with Python and InsightFace, to achieve a precision rate of 93.6%
- Implemented DBSCAN clustering and rigorously verified data accuracy across 182,000+ images, enhancing archive reliability and insight precision.
- Designed an intuitive UI for the labeling tool using Flask and React.js, streamlining user engagement and annotations.
- Designed and maintained database structures using SQL, enhancing user interaction management and feedback data integrity, ensuring operational excellence.

Software Engineer, Ericsson, India **July 2019 – July 2022**

- Developed and maintained enterprise web applications using Java 11, JSP, JSTL, and Spring Framework (Spring MVC, Spring Security, Spring Data), integrating RESTful APIs for backend services. Utilized Apache Maven for dependency management and builds, ensuring consistent deployment across environments.
- Designed dynamic front-end interfaces with HTML5, CSS3, and JavaScript, leveraging React for complex UI components. Improved cross-browser compatibility and accessibility compliance (WCAG 2.0) for government-facing applications.
- Developed MySQL database schemas and optimized SQL queries, achieving a 25% reduction in report generation time. Used JPA with Hibernate for ORM, streamlining data access layers and reducing boilerplate code by 30%.
- Built and maintained SOAP and REST web services using Spring Boot, JAX-WS, and JAX-RS, ensuring seamless integration with legacy systems. Documented APIs with Swagger and conducted load testing using JMeter to validate performance under 10k+ concurrent users.
- Automated CI/CD pipelines using Jenkins, Git, and Docker, reducing deployment time by 40%. Collaborated on containerizing applications for AWS EC2 using Elastic Load Balancer and Auto Scaling, improving system uptime to 99.8%.
- Implemented JUnit and Mockito for unit/integration testing, achieving 95% test coverage. Debugged production issues using Eclipse IDE and SLF4J, resolving critical defects within SLA timelines.
- Migrated monolithic modules to microservices using Spring WebFlow and RabbitMQ for asynchronous communication, reducing service coupling and improving scalability by 35%.
- Authored technical documentation using Confluence and managed tasks via JIRA, aligning with Agile (Scrum) workflows. Conducted bi-weekly code reviews and mentored junior developers on Struts MVC and Grails best practices.
- Generated operational reports using Jaspersoft Studio and automated data pipelines with Automic Automation, reducing manual reporting efforts by 50%.
- Enhanced legacy applications by refactoring XML-based configurations to annotation-driven Spring Boot setups, improving maintainability and reducing startup time by 20%.

TECHNICAL PROJECTS

Sleep data analysis **January 2024**

- Applied LSTM network and Granger causality techniques to investigate sleep data on a dataset comprising over 5,000 individuals' sleep patterns, unveiling insights into temporal dynamics and causal connections within the dataset.